class11

Nicolas Song

Interpreting results

The following is my novel protein from the Find-a-Gene project:

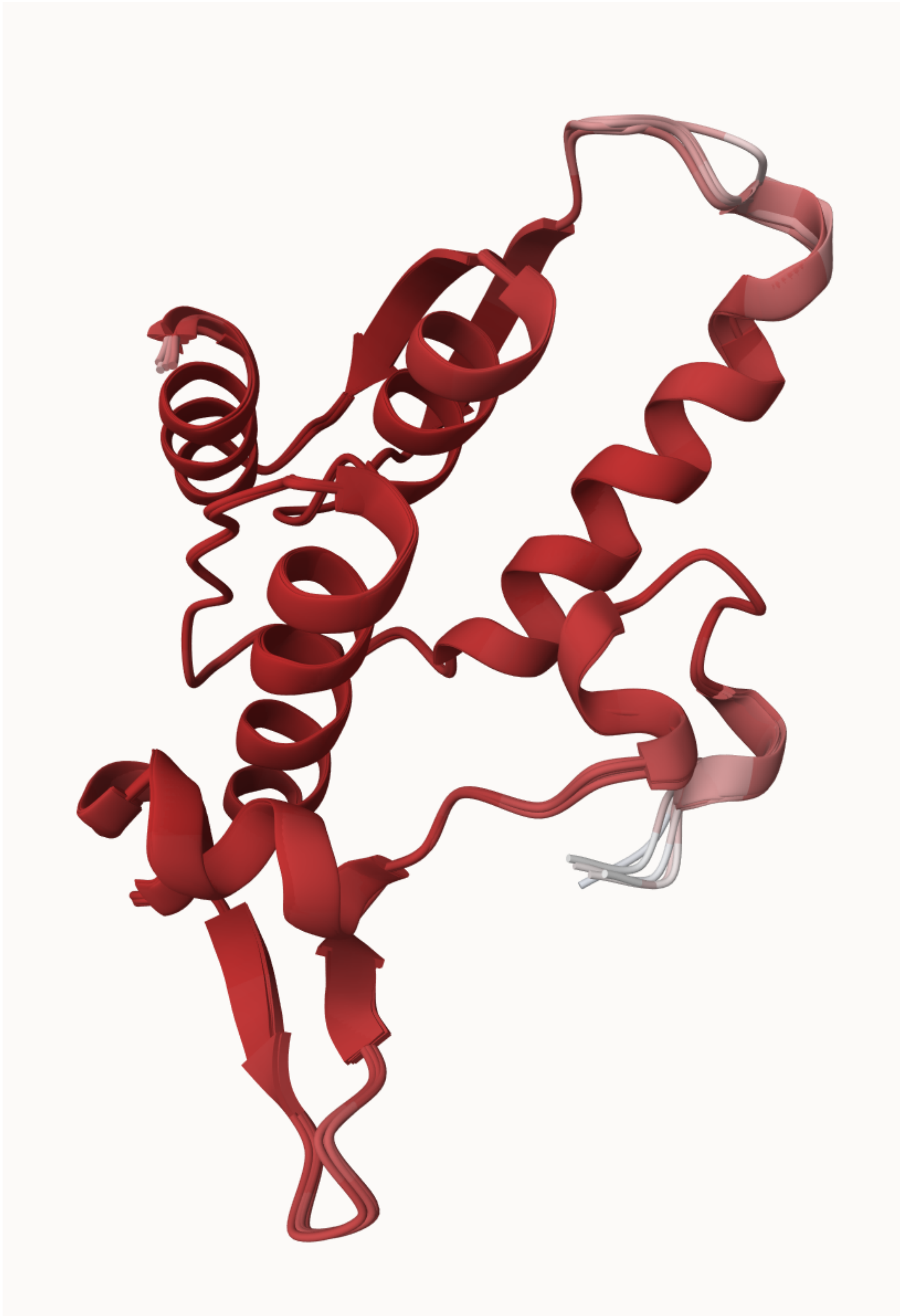


Figure 1: Novel Protein

Custom analysis of resulting models

```r
results_novel <- "find_a_gene_1a78a"

# File names for all PDB models
pdb_files <- list.files(path=results_novel,
    pattern="*.pdb",
    full.names = TRUE)

# Printing PDB file names
basename(pdb_files)
```

```
[1] "find_a_gene_1a78a_unrelaxed_rank_001_alphafold2_ptm_model_5_seed_000.pdb"
[2] "find_a_gene_1a78a_unrelaxed_rank_002_alphafold2_ptm_model_3_seed_000.pdb"
[3] "find_a_gene_1a78a_unrelaxed_rank_003_alphafold2_ptm_model_1_seed_000.pdb"
[4] "find_a_gene_1a78a_unrelaxed_rank_004_alphafold2_ptm_model_4_seed_000.pdb"
[5] "find_a_gene_1a78a_unrelaxed_rank_005_alphafold2_ptm_model_2_seed_000.pdb"
```

```r
library(bio3d)

# Read all data from Models
# and superpose/fit coords
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
find_a_gene_1a78a/find_a_gene_1a78a_unrelaxed_rank_001_alphafold2_ptm_model_5_seed_000.pdb
find_a_gene_1a78a/find_a_gene_1a78a_unrelaxed_rank_002_alphafold2_ptm_model_3_seed_000.pdb
find_a_gene_1a78a/find_a_gene_1a78a_unrelaxed_rank_003_alphafold2_ptm_model_1_seed_000.pdb
find_a_gene_1a78a/find_a_gene_1a78a_unrelaxed_rank_004_alphafold2_ptm_model_4_seed_000.pdb
find_a_gene_1a78a/find_a_gene_1a78a_unrelaxed_rank_005_alphafold2_ptm_model_2_seed_000.pdb
.....

Extracting sequences

pdb/seq: 1   name: find_a_gene_1a78a/find_a_gene_1a78a_unrelaxed_rank_001_alphafold2_ptm_mode
pdb/seq: 2   name: find_a_gene_1a78a/find_a_gene_1a78a_unrelaxed_rank_002_alphafold2_ptm_mode
pdb/seq: 3   name: find_a_gene_1a78a/find_a_gene_1a78a_unrelaxed_rank_003_alphafold2_ptm_mode
pdb/seq: 4   name: find_a_gene_1a78a/find_a_gene_1a78a_unrelaxed_rank_004_alphafold2_ptm_mode
pdb/seq: 5   name: find_a_gene_1a78a/find_a_gene_1a78a_unrelaxed_rank_005_alphafold2_ptm_mode
```

```
pdbs
```

```
                              1        .         .         .         .         50
[Truncated_Name:1]find_a_gen   MALNKFMHPRNLYKEKKPNFQALAVKYPEFRCHTNQEISGKVTLNFKDPD
[Truncated_Name:2]find_a_gen   MALNKFMHPRNLYKEKKPNFQALAVKYPEFRCHTNQEISGKVTLNFKDPD
[Truncated_Name:3]find_a_gen   MALNKFMHPRNLYKEKKPNFQALAVKYPEFRCHTNQEISGKVTLNFKDPD
[Truncated_Name:4]find_a_gen   MALNKFMHPRNLYKEKKPNFQALAVKYPEFRCHTNQEISGKVTLNFKDPD
[Truncated_Name:5]find_a_gen   MALNKFMHPRNLYKEKKPNFQALAVKYPEFRCHTNQEISGKVTLNFKDPD
                               **************************************************
                              1        .         .         .         .         50

                             51        .         .         .         .         100
[Truncated_Name:1]find_a_gen   ALRALSTVLLKEDFGLDVHIPADRLVPTIPLRLNYIHWIEDVLKTTWNDK
[Truncated_Name:2]find_a_gen   ALRALSTVLLKEDFGLDVHIPADRLVPTIPLRLNYIHWIEDVLKTTWNDK
[Truncated_Name:3]find_a_gen   ALRALSTVLLKEDFGLDVHIPADRLVPTIPLRLNYIHWIEDVLKTTWNDK
[Truncated_Name:4]find_a_gen   ALRALSTVLLKEDFGLDVHIPADRLVPTIPLRLNYIHWIEDVLKTTWNDK
[Truncated_Name:5]find_a_gen   ALRALSTVLLKEDFGLDVHIPADRLVPTIPLRLNYIHWIEDVLKTTWNDK
                               **************************************************
                             51        .         .         .         .         100

                            101        .         .         .         .    145
[Truncated_Name:1]find_a_gen   IKGIDIGTGASCIYPLLGSRMNGWKFLASEQDEQSVTSATKNVEQ
[Truncated_Name:2]find_a_gen   IKGIDIGTGASCIYPLLGSRMNGWKFLASEQDEQSVTSATKNVEQ
[Truncated_Name:3]find_a_gen   IKGIDIGTGASCIYPLLGSRMNGWKFLASEQDEQSVTSATKNVEQ
[Truncated_Name:4]find_a_gen   IKGIDIGTGASCIYPLLGSRMNGWKFLASEQDEQSVTSATKNVEQ
[Truncated_Name:5]find_a_gen   IKGIDIGTGASCIYPLLGSRMNGWKFLASEQDEQSVTSATKNVEQ
                               *********************************************
                            101        .         .         .         .    145

Call:
  pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
  5 sequence rows; 145 position columns (145 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

```r
# Calculate RMSD between all pairs models
rd <- rmsd(pdbs, fit=T)
```

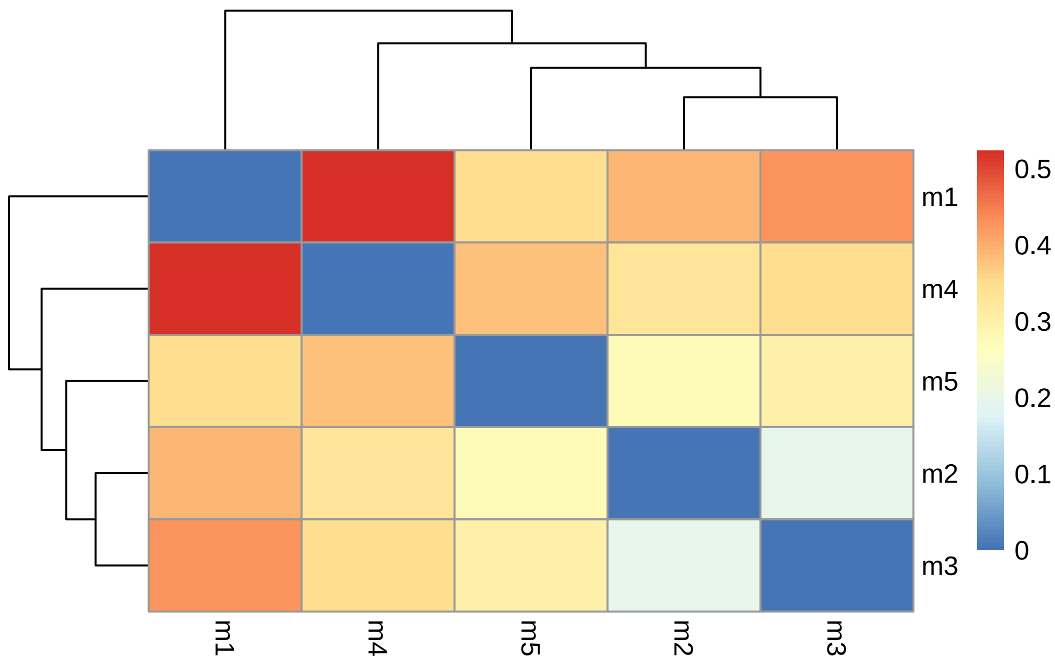
```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 145 non NA positions
```

```r
range(rd)
```

```
[1] 0.000 0.524
```

```r
# Drawing a heatmap of RMSD matrix values
library(pheatmap)

colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```




```r
# Read a reference PDB structure
pdb <- read.pdb("6b91")
```

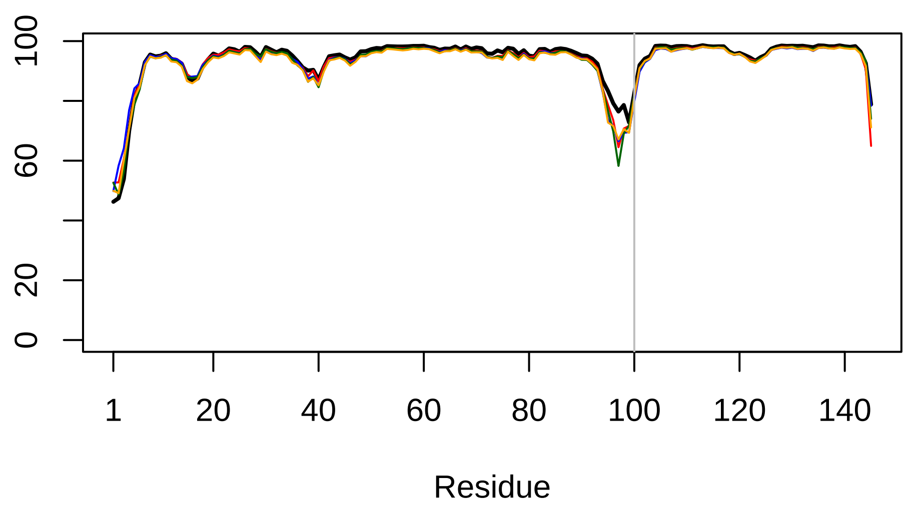
```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

Used most similar known protein, human METTL16

```r
# Plotting pLDDT values across all models
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
```

```
Warning in plotb3(pdbs$b[1, ], typ = "l", lwd = 2, sse = pdb): Length of input
'sse' does not equal the length of input 'x'; Ignoring 'sse'
```

```r
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```




```r
core <- core.find(pdbs)
```

```
 core size 144 of 145  vol = 0.325
 FINISHED: Min vol ( 0.5 ) reached
```

```r
core.inds <- print(core, vol=0.5)
```

```
# 145 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     1 145    145
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

When corefit_structures opened in Mol*, the following structure (already superimposed) was shown:

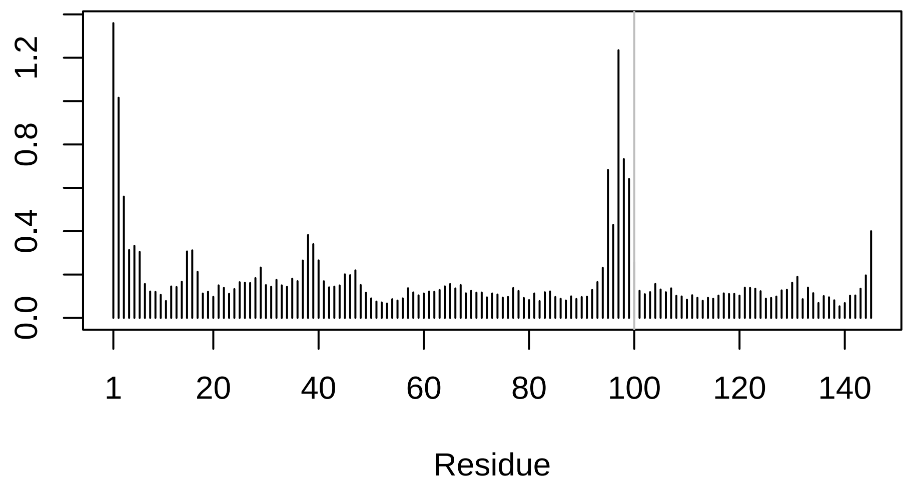
Figure 2: Core Fit Structure

```r
# Examining RMSF between positions of the structure
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
```

```
Warning in plotb3(rf, sse = pdb): Length of input 'sse' does not equal the
length of input 'x'; Ignoring 'sse'
```

```r
abline(v=100, col="gray", ylab="RMSF")
```



Predicted Alignment Error for domains

```r
library(jsonlite)

# Listing of all PAE JSON files
pae_files <- list.files(path=results_novel,
    pattern=".*model.*\\.json",
    full.names = TRUE)
```

```r
# Reading 1st file
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
# Reading 5th file
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"
```

```r
# Per-residue pLDDT scores
#  same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 46.25 47.44 53.88 69.94 80.88 85.56
```

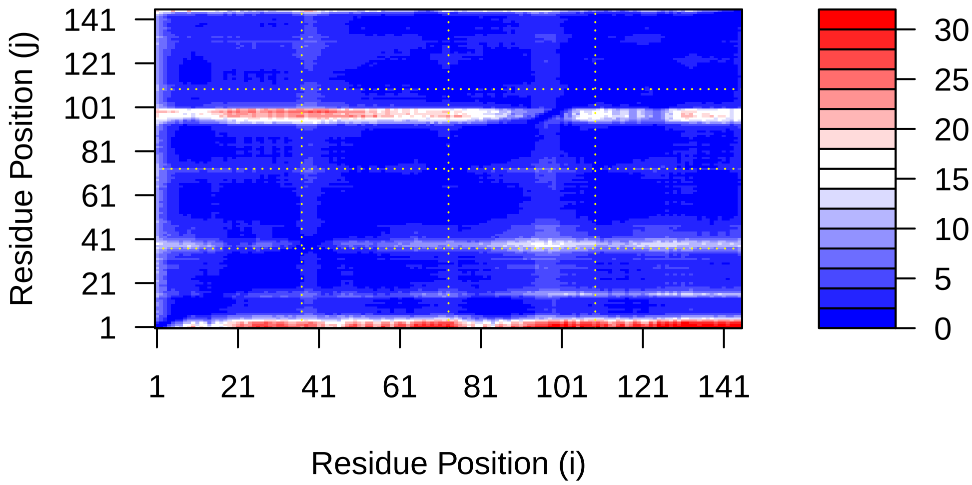
```r
# Max PAE value for model 1
pae1$max_pae
```

```
[1] 30.67188
```

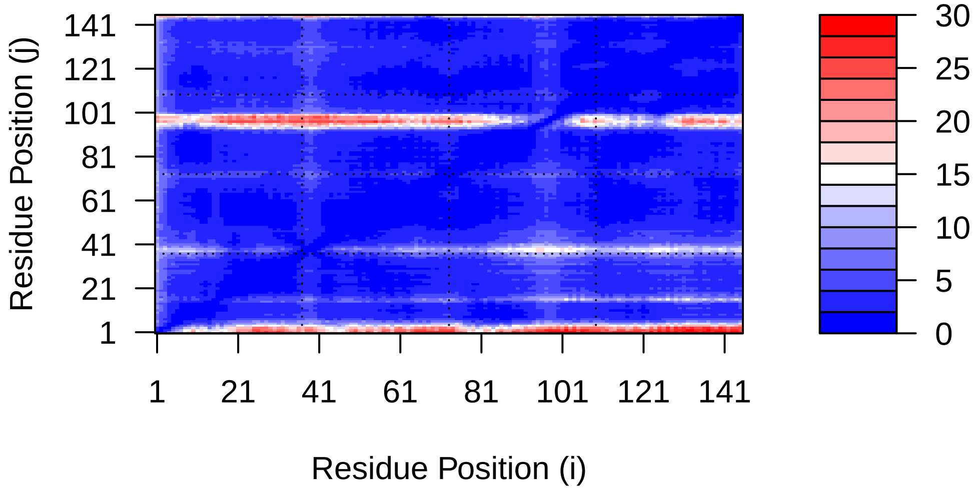
```r
# Max PAE value for model 5
pae5$max_pae
```

```
[1] 29.67188
```

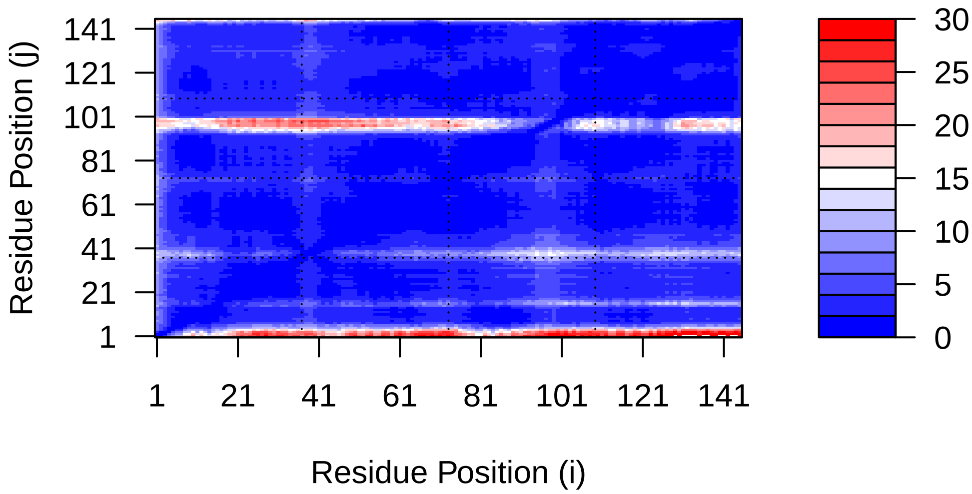
```r
# Plot PAE scores for model 1
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```




```r
# Plot PAE scores for model 5
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```




```r
# Model 1 PAE plot with same data range as model plot
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Residue conservation from alignment file

```
aln_file <- list.files(path=results_novel,
      pattern=".a3m$",
      full.names = TRUE)
aln_file
```

```
[1] "find_a_gene_1a78a/find_a_gene_1a78a.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

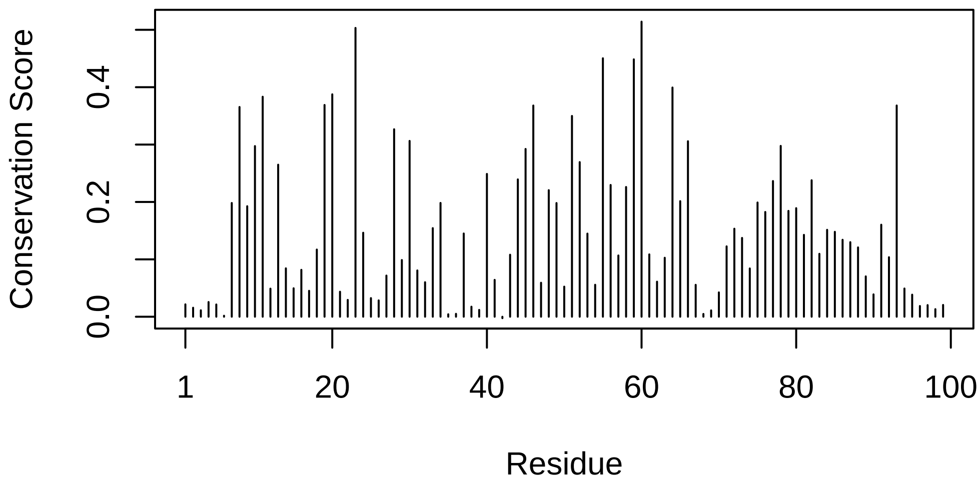
```
[1] " ** Duplicated sequence id's: 101 **"
```

```
# Seeing how many sequences
dim(aln$ali)
```

```
[1] 6133  338
```

```r
# Scoring residue conservation
sim <- conserv(aln)
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```

```
Warning in plotb3(sim[1:99], sse = trim.pdb(pdb, chain = "A"), ylab =
"Conservation Score"): Length of input 'sse' does not equal the length of input
'x'; Ignoring 'sse'
```




```r
# Consensus sequence
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
```

```
[145] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[163] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[181] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[199] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[217] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[235] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[253] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[271] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[289] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[307] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[325] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
```

```r
m1.pdb <- read.pdb(pdb_files[1])
# Assigning a random alignment (lengths not equal)
occ <- vec2resno(sim[1:145], m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```

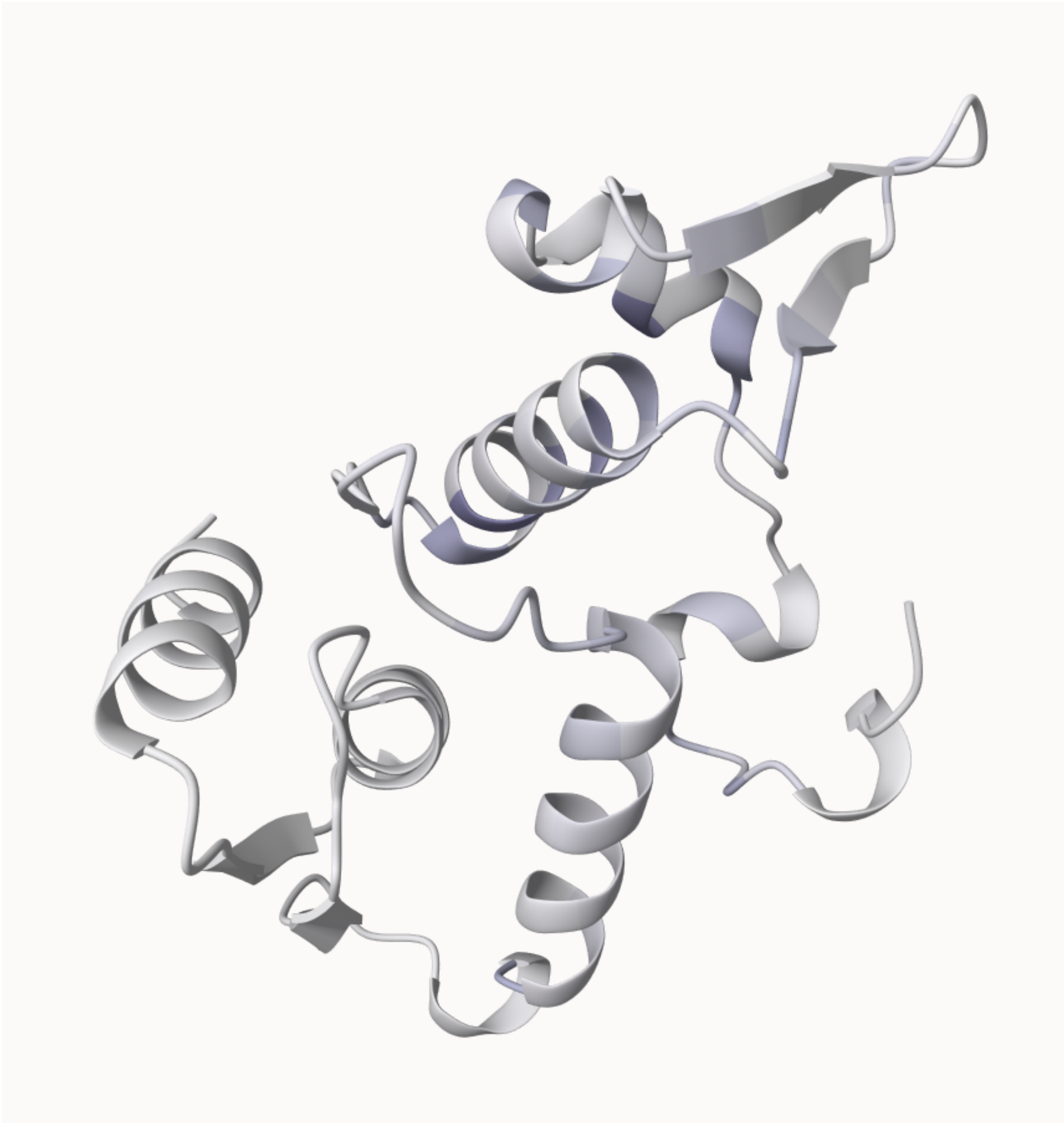


Figure 3: M1 Conserved PDB